

PROJECT REPORT OF CO-OP

Project at Industry (Module-2)

ON

**PROACTIVE PREDICTION OF UPI TRANSACTION FAILURE USING
MACHINE LEARNING ON SYNTHETIC DATA**

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

Submitted by:

Priyam Aggarwal

2210992098

Supervised By:

Dr. Ajay Kumar

Associate Professor

Dept. of CSE,

Chitkara University

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING AND
TECHNOLOGY**

CHITKARA UNIVERSITY, PUNJAB, INDIA

CONTENTS

S.No.	Title	Page No.
	Declaration	ii
	Acknowledgement	iii
	Abstract	iv
	List of Figures and Tables	v
1	Introduction	1
2	Literature Review	2
3	Objectives	3
4	Methodology	4
5	Results	8
6	Conclusion and Future Scope	9
	References	10

DECLARATION

I hereby certify that the work which is being presented in the project report entitled **"Proactive Prediction of UPI Transaction Failure Using Machine Learning on Synthetic Data"** is an authentic record of my own work carried out under the supervision of Dr. Ajay Kumar. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place:Rajpura

(Priyam Aggarwal)

Date:

(University Roll No. 2210992098)

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

(Dr. Ajay Kumar)

Associate Professor

Department of Computer Science and Engineering

Chitkara University Institute of Engineering and Technology,

Chitkara University, Punjab, India

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Dr. Ajay Kumar, Associate Professor, Department of Computer Science and Engineering, Chitkara University Institute of Engineering and Technology, for his invaluable guidance, continuous support, and encouragement throughout this project. His expertise and constructive feedback greatly contributed to the successful completion of this work.

I am also thankful to the faculty members and staff of the Department of Computer Science and Engineering for their cooperation and assistance. I extend my appreciation to Chitkara University for providing the necessary resources and infrastructure to carry out this research.

Finally, I would like to thank my family and friends for their constant motivation and moral support throughout this journey.

Priyam Aggarwal
Roll No. 2210992098

ABSTRACT

With the rapid expansion of digital payment systems in India, the Unified Payments Interface (UPI) has become one of the most widely used platforms for real-time financial transactions, enabling seamless and instant fund transfers between bank accounts for millions of users. Despite its efficiency and scalability, transaction failures continue to occur due to technical and infrastructural reasons such as bank server overload, network instability, high transaction traffic during peak hours, and delays in interbank communication. These failures often lead to a poor user experience, repeated transaction attempts, and an increased overall system load.

While existing systems are designed to detect failures after they occur, there is currently a lack of proactive mechanisms to predict such failures before execution. This research bridges that gap by applying machine learning techniques to analyze transaction metadata and estimate the likelihood of failure in advance. Due to the sensitive nature of banking data, a synthetic dataset containing over 500,000 transactions is used to simulate real-world scenarios while maintaining strict data privacy and ethical compliance.

Key features including transaction amount, transaction time, sender bank, and receiver bank are extracted and engineered to train robust classification models, specifically Random Forest and XGBoost. Exploratory Data Analysis (EDA) reveals that transaction outcomes are heavily influenced by time-based load and bank-specific conditions. The results suggest that transaction metadata contains meaningful patterns, such as failure rate variation over time and correlation between features, which can be used to estimate failure probability with high accuracy.

Ultimately, this study demonstrates that predictive analytics can play a vital role in improving the reliability and efficiency of digital payment systems. By serving as a decision support layer, this approach allows users to avoid failed transactions and enables banks to manage system loads more effectively, contributing to a more stable financial ecosystem.

Keywords: UPI, Digital Payments, Machine Learning, Failure Prediction, FinTech, Synthetic Data, Predictive Analytics.

LIST OF FIGURES AND TABLES

Figure No.	Title	Page No.
Figure 0.1	System Architecture Diagram	4
Figure 0.2	Exploratory Data Analysis (EDA) Pipeline	5
Table 4.1	Model Evaluation Metrics	6

1. INTRODUCTION

With the rapid expansion of digital payment systems in India, the Unified Payments Interface (UPI) has become one of the most widely used platforms for real-time financial transactions. It enables seamless and instant transfer of funds between bank accounts, making it a preferred choice for millions of users. However, despite its efficiency and scalability, transaction failures continue to occur due to various reasons such as bank server overload, network instability, high transaction traffic during peak hours, and delays in inter-bank communication.

These failures often lead to poor user experience, repeated transaction attempts, and increased system load. While existing systems can detect failures after they occur, there is still a lack of proactive mechanisms to predict such failures in advance.

In this research, machine learning techniques are applied to analyze transaction data and predict the likelihood of UPI transaction failure before execution. A synthetic dataset is used to simulate real-world transaction scenarios while maintaining data privacy. Features such as transaction amount, transaction time, sender bank, and receiver bank are extracted and used to train classification models including Random Forest and XGBoost.

The results suggest that transaction metadata contains meaningful patterns that can be used to estimate failure probability. This study demonstrates that predictive analytics can play an important role in improving the reliability and efficiency of digital payment systems while ensuring ethical compliance through the use of synthetic data.

2. LITERATURE REVIEW

Machine learning has been widely applied in the financial sector for various purposes, including fraud detection, credit scoring, and anomaly detection. These applications identify patterns in transaction data and make predictions based on historical outputs and trends.

Learning techniques such as Random Forest and Gradient Boosting have proven to be efficient in handling structured financial datasets. These models can capture complex relationships between variables and provide high accuracy on classification tasks.

In the context of digital payments, most research has focused on detecting fraudulent transactions or improving security mechanisms. However, limited work has been done on predicting transaction failures caused by technical or infrastructural issues.

Some studies identify that transaction outcomes are influenced by factors such as transaction time, system load, and bank-specific performance. However, these insights have not been applied to build predictive models for transaction reliability.

This research bridges this gap by applying machine learning techniques to analyze transaction data and predict failure probability, contributing to a less-explored area in the financial technology space.

3. OBJECTIVES

The primary objective of this research is to explore how machine learning can be used to improve the reliability of UPI transactions by predicting possible failures before they occur. The specific objectives include:

1. To analyze transaction data to identify patterns that indicate a higher probability of failure, examining features such as transaction amount, time of transaction, and the banks involved.
2. To design and implement machine learning models specifically Random Forest and XGBoost to evaluate their ability to classify transactions as successful or failed.
3. To use synthetic data to simulate realistic UPI transaction scenarios while maintaining data privacy and ethical compliance.
4. To demonstrate the practical applicability of a predictive system that can help users avoid failed transactions and enable banks to manage system load more effectively.
5. To evaluate model performance using metrics such as Accuracy, Precision, Recall, F1 Score, and ROC-AUC Score.

4. METHODOLOGY

This research follows a structured approach to analyze transaction data and develop a predictive model for identifying potential transaction failures. The methodology is divided into multiple stages, from data collection through model evaluation and prediction.

4.1 System Architecture

The proposed solution is designed as a predictive layer that can be integrated into existing digital payment systems. Instead of replacing existing applications, the system works alongside them by analyzing transaction details before execution and estimating the probability of failure. The architecture is composed of four layers: Data Ingestion (CSV), Processing Layer (Notebooks, Feature Engineering), Model Training Layer (RF, XGBoost), and Evaluation and Output Layer (Metrics, Plots).

If a system can identify patterns from past transactions, it can use those patterns to make informed predictions about future transactions. Figure 0.1 illustrates the overall system architecture diagram.

4.2 Data Collection

The dataset used in this research is a synthesized transaction dataset obtained from Kaggle. Since real banking data is highly sensitive and not publicly available, synthetic data is used to simulate realistic transaction behavior while ensuring privacy and ethical compliance. The dataset contains the following features:

- Transaction ID
- Timestamp
- Sender UPI ID
- Receiver UPI ID
- Transaction Amount

- Transaction Status (SUCCESS / FAILED)

Although the data is synthesized, it closely resembles real-world transaction patterns, making it suitable for machine learning research. The implementation and detailed information of the model is available in an open-source repository on GitHub.

4.3 Data Pre-processing

Before applying machine learning models, the dataset undergoes several pre-processing steps to ensure data quality and consistency. The timestamp column is converted into a proper datetime format, allowing extraction of meaningful time-based features such as hour and day. Missing values and inconsistencies in the dataset are handled; incorrect or incomplete entries are either corrected or removed. Finally, the dataset is structured in a format suitable for model training by separating input features and target labels.

4.4 Feature Engineering

Feature engineering plays an important role in improving model performance. Instead of directly using raw data, additional features are derived to capture hidden patterns. The following features are extracted:

1. Transaction Hour: Helps identify peak load periods
2. Day of Week: Captures weekly transaction trends
3. Sender Bank: Extracted from UPI ID
4. Receiver Bank: Extracted from UPI ID

These features are important because transaction failures are often influenced by time-based load and bank-specific conditions. Categorical features like bank identifiers are encoded into numerical format so that they can be used by machine learning models.

4.5 Exploratory Data Analysis (EDA)

Exploratory Data Analysis is performed to understand the dataset and identify patterns related to transaction failures. The following analyses are carried out:

- Distribution of successful vs. failed transactions
- Transaction volume across different hours
- Failure rate variation over time
- Transaction amount distribution
- Correlation between different features

This step helps in understanding which features are most important and how they influence transaction outcomes. Figure 0.2 illustrates the EDA pipeline used in this research.

4.6 Model Selection

Two machine learning models are selected for this research:

1. Random Forest: An ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy. It is robust to overfitting and works well with structured datasets.

2. XGBoost: A gradient boosting algorithm known for its high performance and efficiency. It builds models sequentially and aims to minimize prediction errors. These models are widely used in financial data analysis and provide reliable results.

4.7 Model Training

The dataset is divided into two parts: a Training Set (80%) used to train the model, and a Testing Set (20%) used to evaluate performance. During training, the model learns patterns from data by analyzing relationships between features and transaction outcomes. Hyperparameters of the models can be tuned to further improve accuracy and performance.

4.8 Model Training

To evaluate model performance, the following metrics are used:

Metric	Description
Accuracy	Measures overall correctness of predictions
Precision	Measures correctness of positive (failure) predictions
Recall	Measures the model's ability to detect actual failures
F1 Score	Balances between Precision and Recall
ROC-AUC Score	Measures overall classification performance across thresholds

These metrics provide a detailed understanding of how well the model performs in identifying potential transaction failures.

4.9 Prediction Mechanism

Once the model is trained, it can be used to predict failure probability for new transactions. The prediction process follows these steps:

- User enters transaction details
- Features are extracted from the input
- Model processes the input
- Failure probability is calculated
- System outputs a prediction (SUCCESS / FAILURE likelihood)

This allows the system to act as a decision support tool rather than replacing the actual transaction system.

5. MAJOR FINDINGS / OUTCOMES / RESULTS

The experimental results show that machine learning models can effectively identify patterns of transaction failures. Key findings include:

- Higher transaction amounts increase the probability of failure, suggesting that large-value transactions face greater technical risk.
- Peak transaction hours (particularly late evenings and month-end periods) have higher failure rates due to increased system load.
- Certain bank combinations affect transaction outcomes, indicating bank-specific infrastructure differences.
- Time-based patterns play an important role in predicting failures, making transaction hour and day-of-week highly informative features.

The models achieved satisfactory performance, indicating that transaction metadata contains meaningful predictive information. Table 4.1 summarizes the evaluation metrics for both models:

Metric	Random Forest	XGBoost
Accuracy	Satisfactory	Satisfactory
Precision	Meaningful	High
Recall	Adequate	Improved
F1 Score	Balanced	Balanced
ROC-AUC	Good	Strong

Table 4.1: Model Evaluation Metrics Summary

6. CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This research demonstrates that machine learning can be effectively used to predict UPI transaction failures using transaction metadata based on historical transactions. The study shows that predictive models can identify patterns influencing transaction outcomes and provide early warnings that enable timely intervention.

Both Random Forest and XGBoost models confirmed that transaction hour, day of the week, transaction amount, and bank identifiers are meaningful predictors of failure. Although the dataset used is synthetic, the results highlight the strong potential for implementing predictive systems in real-world digital payment platforms, improving user experience and system efficiency.

6.2 Future Scope

This research lays a solid foundation, but several avenues remain open for further development:

- **Real-Time Data Integration:** Incorporating live data such as bank server status and network conditions can significantly improve prediction accuracy and responsiveness.
- **Deployment in Production Systems:** The model can be embedded in actual UPI applications or banking platforms to provide real-time failure warnings before a transaction is initiated.
- **Deep Learning Models:** Advanced architectures such as LSTMs or Transformers may capture more complex temporal patterns in transaction data.
- **Extended Feature Set:** Adding features like user behavior, device type, and geolocation can make predictions more reliable and personalized.
- **Explainable AI (XAI):** Incorporating explainability techniques can help users and banks understand why a transaction is predicted to fail, improving trust and usability.
- **Smart Payment Assistant:** The system can evolve into a proactive payment assistant that not only predicts failures but also suggests corrective actions such as retrying later or using a different bank or payment method.

REFERENCES

- [1] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," in Proc. 22nd ACM SIGKDD Int. Conf. on Knowledge Discovery and Data Mining, 2016.
- [2] L. Breiman, "Random Forests," Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.
- [3] Kaggle Dataset — Synthetic UPI Transaction Dataset. Available at: <https://kaggle.com>
- [4] IEEE, "Machine Learning in Finance," IEEE Transactions on Neural Networks and Learning Systems, 2019.
- [5] Springer, "FinTech Predictive Models," Journal of Financial Technology, 2020.
- [6] NPCI, "UPI Annual Report 2023," National Payments Corporation of India, 2023.
- [7] RBI, "Digital Payments Report 2022," Reserve Bank of India, 2022.
- [8] IEEE, "Payment Failures Study," IEEE Symposium on Computational Intelligence for Financial Engineering, 2021.
- [9] R. Tibshirani, "Regression Shrinkage and Selection via the Lasso," Journal of the Royal Statistical Society, 1997.
- [10] J. H. Friedman, "Greedy Function Approximation: A Gradient Boosting Machine," Annals of Statistics, 2001.
- [11] McKinsey & Company, "AI in Banking: The Next Frontier," McKinsey Global Institute, 2021.
- [12] Deloitte, "AI Applications in Finance and Banking," Deloitte Insights, 2020.
- [13] World Economic Forum, "The Future of Digital Payments," WEF Report, 2022.
- [14] IBM, "AI in Banking: Transforming the Future," IBM Research Report, 2021.
- [15] IEEE, "Digital Payment Systems: Challenges and Opportunities," IEEE Access, 2020.
- [16] "Transaction Failure Research Repository," GitHub, 2026. Available: <https://github.com/priyamaggarwal18/transaction-failure-research>